

Pre-PGI Process, Callable Surface and the **One-API Decision**

What SAP actually does between delivery and goods issue, which parts of it can be called, and why the pre-PGI span should be a single interface.

SYSTEMS	DATE	WRITES PERFORMED	COMPANION
DS4 / 200 QS4 / 700	18 Aug 2026	None	Stakeholder brief (sections align 1:1)

THE FINDING THAT DECIDES IT

Four of the six pre-PGI steps have **no callable SAP interface at all**. Custom development is mandatory either way — the only question is how many doors we cut.

Every figure and document number in this document is a live system read. Nothing is estimated or inferred from documentation.

How to use this. The stakeholder brief carries the same section numbers with less depth. Read this one fully; present that one. If challenged on any claim in the brief, the supporting detail is in the matching section here.

Contents

1 How this was established

Access, evidence standard, tools

2 What SAP actually does before PGI

Six states, a real chain, status semantics, document linkage

3 What is callable, and what is not

OData services, BAPIs, internal modules, the verdict

4 What we found that is not currently known

10,496 stranded · cost type · the PGI gate · existing implementation

5 Comparison with the v1.8 specification

Corrections, the \$4 gap, API-by-API status

6 The decision: one pre-PGI API

The question, four arguments, the contract, three conditions

7 What remains open

Unproven items, the DS4 blocker, what we need

8 Quick reference

Document numbers, sentences to use, traps to avoid

1 How this was established

1.1 Systems, access and what was touched

System	Client	User	Access	Writes performed
DS4 — development	200	QNOVATE8	Write-authorised	None — all reads
QS4 — quality	700	QNOVATE8	Strictly read-only	None

No SAP document was created, changed, posted, committed or reversed in either system. Every finding comes from reading metadata, function-module interfaces, repository indexes and existing document tables.

Why two systems

DS4 is where we are authorised to write, so it is where runtime certification must eventually happen — but it has almost no business data, which is itself a finding (see 7.2). QS4 has millions of real documents but is read-only. The pattern throughout is therefore: **learn the mechanism in DS4** (metadata, interfaces, repository indexes), **observe the behaviour in QS4** (real documents, real statuses).

1.2 What counts as evidence

This distinction is the difference between “we think” and “we know”, and it is what will be tested in the room.

Counts as evidence	Does <i>not</i> count
A live <code>\$metadata</code> response with status captured	A service appearing in a catalogue or SEGW list
An SE37 interface read from the live system	A payload generated from that interface
A where-used result <i>with a control</i> proving the index covers the object type	A where-used result with no control — scope silently excludes types
An SE16 hit count from the result title	A row sample from a lazy-loading grid treated as population-scale
A document chain traced field by field	Two close timestamps inferred to be one transaction

TERMS THAT GET CONFUSED — AND THE DIFFERENCE CHANGES THE ANSWER

Remote-enabled ≠ released. `TFDIR-FMODE = R` means callable over RFC. Release status is a separate attribute meaning SAP commits to interface stability across upgrades. Three modules here are remote-enabled but *not* released — adopting them is a supportability decision.

HTTP 200 ≠ business authorisation. A `$metadata` 200 proves registration, an active ICF node and a resolving alias. It proves nothing about whether the business operation will succeed.

Empty result ≠ no data exists. It means no rows matched *that request* under *that user's* authorisations.

1.3 The tools, and what each one told us

Tool	What it answers	Used here for
<code>/IWFND/GW_CLIENT</code>	Is this OData service live, and what is its exact contract?	<code>\$metadata</code> for all seven services; filtered GETs
<code>/IWFND/MAINT_SERVICE</code>	Registered, with which alias and ICF state?	Confirmed local registration under <code>\$TMP</code>
SEGW	What design-time model exists?	The 2,626-project catalogue v1.8 was built from
SE37	Real interface, release status, RFC flag	15 BAPI interfaces + the entire <code>SD_SCD*</code> family; two BAPIs executed
Where-Used (SE37 / SE24)	Who actually calls this?	The decisive tool — found the internal family and proved who calls it
SE16	What does the data actually look like?	Status distributions, document chains, config tables, the 10,496 count
<code>DD03L</code> / <code>DD07L</code> / <code>DD07T</code>	What does this field <i>mean</i> ?	Corrected the project's single biggest misunderstanding — see 2.3

2 What SAP actually does before PGI

2.1 The six states

The portal presents one action. SAP performs six independently committed state transitions. Each is a separate database commit; none can be undone by a rollback once committed.

#	State	Stored in	Proof it completed
1	Shipment created	VTTK header, VTTT assignment	TKNUM exists
2	Cost document created	VFKK header, VFKP item	FKNUM exists
3	Calculated	VFKP	STBER = C + DTBER / UZBER
4	Account assigned	VFKP	STFRE = C + DTFRE / UZFRE
5	Settled / transferred	VFKP + MM	STABR = C, EBELN + LBLNI populated
6	Goods issue (PGI)	MKPF / MSEG, LIKP	Material doc, movement 601; WBSTK = C

The shipment-level mirror

The shipment header carries its own summary of steps 3 and 5, which lets you assess a shipment without joining to the cost tables:

- VTTK-FBGST — shipment cost **calculation** status
- VTTK-ARGST — shipment cost **settlement** status

Shipment 1100605471 : both C . Shipment 2100005093 : FBGST = C , ARGST absent — calculated, never settled.

2.2 A real completed chain, end to end

Delivery 9004953084 , QS4, 17 August 2026. This is the reference example — these numbers recur throughout both documents.

10:26:23	●	Delivery 9004953084 created Type ZLF · shipping point 7683 · ship-to 0020105990 · route E36881
11:11:28	●	Shipment 1100605471 created Type Z001 · planning point 7683 · carrier 0013000730 · 484 KM · ZZVEHICLE_NO HR55AZ3800
11:11:28	●	Cost document 1100608871 created, calculated, account assigned FKART Z001 · 7,460.00 INR
11:11:30	●	Cost settled — transferred to MM Freight PO 6000105831 · service entry 1003382363
11:11:31	●	PGI material document 4918168265 Movement 601
11:11:32	●	Billing 1108024764 VBTP M
12:33-14:48	●	Five cancel / re-invoice cycles VBTP N → M · outside pre-PGI scope, but real API-07 evidence

TWO SECONDS

Shipment created to cost settled took 11:11:28 → 11:11:30. This is not a long-running process needing external checkpointing between steps. It is a fast sequential chain — which matters directly for the decision in section 6.

2.3 Status fields — the correction that changes the project's vocabulary

MOST IMPORTANT CORRECTION IN THIS DOCUMENT

The project has treated VFKK-STFRE as the “cost release” field for weeks. It is not. It is the account assignment status.

Read from the data dictionary, not from field names. Domain texts via DD07T :

Field	Domain	Actual meaning	blank	A	B	C
STBER	STBER_K	Calculation	not relevant	not calculated	partial	fully calculated
STFRE	STFRE_K	Account assignment	not relevant	not completed	partial	completed
STABR	STABR_K	Transfer / settlement	not relevant	not transferred	partial	fully transferred

Three consequences, each of which has already burned us

- STFRE is not release. The name resembles German Freigabe (release); the domain text says Kontierung (account assignment). Field names lie; domains do not.
- Blank means “not relevant”, not “not done”. An earlier analysis found 999 of 999 sampled documents blank on STFRE and concluded release never happens. Invalid — blank meant those documents were never supposed to be account-assigned.
- VFKK has no release field at all. It has exactly 18 fields. What the business calls “cost release” is the transfer step, STABR — whose function module is literally named for it (see 3.3).

2.4 How the documents link — and a join that was wrong

CORRECTION

Earlier project notes recorded FKNUM = REBEL = TKNUM — cost document number equals shipment number. This is false.

Cost doc	FKNUM	VFKP-REBEL	Actual shipment
1100608871		1100605471	1100605471
2100005093		2100005099	2100005099 — not 2100005093
1300105310		1300105363	1300105363

The correct join is VFKP-REBEL → VTTK-TKNUM . The number ranges overlap closely enough that the two look interchangeable — exactly how the error got in. Any reporting or reconciliation logic keyed on FKNUM = TKNUM will silently produce wrong results.

Full linkage path: VFKK/VFKP → REBEL → VTTK → VTPP → VBELN → LIKP/LIPS → VBFA → material document and billing.

3 What is callable, and what is not

3.1 The OData services

Seven services registered and live in DS4/200. All returned HTTP 200 for `$metadata` through `/IWFND/GW_CLIENT`. Host `vhresds4ci.sap.shreecement.com:44300`. Registered locally under package `$TMP` — acceptable for development, **not** a transportable deployment.

Service	Ver	Metadata	Serves
<code>API_MATERIAL_DOCUMENT_SRV</code>	1	40,835 B	API-01 goods receipt
<code>API_OUTBOUND_DELIVERY_SRV</code>	2	159,228 B	API-02, API-03 picking/PGI, API-09, API-12
<code>API_BILLING_DOCUMENT_SRV</code>	1	143,849 B	API-03 read-back, API-10
<code>API_PURCHASEORDER_PROCESS_SRV</code>	1	119,171 B	API-08, API-11
<code>API_MATERIAL_STOCK_SRV</code>	1	16,608 B	API-05 book stock
<code>MMIM_MATDOC_SRV</code>	1	71,809 B	API-01 read-back companion
<code>SD_CUSTOMER_INVOICES_CREATE</code>	1	225,238 B	API-03 billing create (application-internal)

Parsed from those responses: 1,854 properties · 110 entity sets · 23 function imports · 81 navigation properties.

THE DELIVERY SERVICE MUST BE VERSION 2

Exact runtime path: `/sap/opu/odata/sap/API_OUTBOUND_DELIVERY_SRV;v=2/`, generated from SEGW project `API_OUTBOUND_DELIVERY_0002`. Version 1 exists under the identical technical name and **lacks the batch-split operations**. It would activate cleanly and then fail on first real use.

Write capability, from the service's own annotations

Entity set	Create	Update	Delete	Meaning
<code>A_MaterialDocumentHeader</code>	yes	false	false	Deep insert from header only
<code>A_MaterialDocumentItem</code>	false	false	false	Items cannot be posted standalone
<code>A_OutbDeliveryHeader</code>	yes	yes	yes	API-02 create supported
<code>A_OutbDeliveryItem</code>	false	yes	yes	API-12 PATCH supported; no standalone create
<code>A_PurchaseOrder</code> + item + schedule	yes	yes	yes	API-11 deep create supported
<code>A_MaterialStock</code> , <code>A_MatLStkInAcctMod</code>	false	false	false	API-05 read-only, as required
All 8 <code>A_BillingDocument*</code> sets	false	false	false	Billing cannot be created here

LOAD-BEARING

`API_BILLING_DOCUMENT_SRV` is read-only apart from its `Cancel` function import. Billing creation must go through `SD_CUSTOMER_INVOICES_CREATE` or `BAPI_BILLINGDOC_CREATEMULTIPLE`. Any design assuming the billing API can create is wrong.

Delivery v2 function imports — exact runtime signatures

All are POST (`m:HttpMethod` from live metadata, not inferred from the name). The only GET in the entire set is billing `GetPDF`.

Function import	Params	Signature
PostGoodsIssue	1	DeliveryDocument
ReverseGoodsIssue	2	+ ActualGoodsMovementDate
ConfirmPickingAllItems	1	DeliveryDocument
ConfirmPickingOneItem	2	+ DeliveryDocumentItem
PickAllItems / PickOneItem	1 / 2	document, item
PickOneItemWithBaseQuantity	4	+ ActualDeliveredQtyInBaseUnit , BaseUnit
PickOneItemWithSalesQuantity	4	+ ActualDeliveryQuantity , DeliveryQuantityUnit
SetPickingQuantityWithBaseQuantity	4	base qty + unit
PickAndBatchSplitOneItem	5	+ Batch , SplitQuantity , SplitQuantityUnit
CreateBatchSplitItem	6	+ PickQuantityInSalesUOM , ActualDeliveryQuantity , DeliveryQuantityUnit

Both bolded counts correct v1.8, which listed four parameters for **PickAndBatchSplitOneItem**. **Not yet proven:** parameter encoding — OData V2 POST function imports conventionally use the query string, but no executed call has confirmed it here.

3.2 The BAPIs

Fifteen interfaces captured live from SE37 in DS4/200. All remote-enabled. **Three are not released.**

BAPI	Released	Role in the flow
BAPI_SHIPMENT_CREATE	Yes	The one pre-PGI step with a proper standard API
BAPI_SHIPMENT_CHANGE	No	Change shipment; <i>not</i> proven to perform cost release
BAPI_SHIPMENT_COST_ESTIMATE	No	Estimate only — creates no document
BAPI_GOODSMVT_CREATE / _CANCEL	Yes	API-01 receipt / reversal
BAPI_OUTB_DELIVERY_CREATE_STO / _SLS	Yes	API-02 alternatives
BAPI_OUTB_DELIVERY_CHANGE	No	API-12 alternative
BAPI_OUTB_DELIVERY_CONFIRM_DEC	Yes	Picking confirmation
BAPI_PO_CREATE1	Yes	API-11
BAPI_BILLINGDOC_CREATEMULTIPLE / _CANCEL1	Yes	Billing
BAPI_MATERIAL_AVAILABILITY	Yes	ATP — not book stock
BAPI_TRANSACTION_COMMIT / _ROLLBACK	Yes	LUW control

BAPI_SHIPMENT_CREATE — the input surface

Imports: HEADERDATA (BAPISHIPMENTHEADER), ITEMDATA , STAGEDATA , HEADERDEADLINE , STAGEDEADLINE , ADDRESS , HDUNHEADER , HDUNITEM , ITEMONSTAGE . Exports: TRANSPORT (shipment number), SHIPMENTGUID , RETURN .

THE CLIENT'S TRANSPORT FIELDS LIVE HERE

BAPISHIPMENTHEADER carries the customer append fields ZZVEHICLE_NO , ZZDRIVER_NAME , ZZDRIVERMOB , ZZLR_GR_NO , ZZLR_GR_DATE , ZZGROSS_WT , ZZNET_WT , ZZTARE_WT . v1.8 §8 parked these as “belonging to the unidentified Stage-B route”. **They are now placed** — the shipment-create call is their request surface.

NO SIMULATION MODE

There is no **TESTRUN** parameter. The first execution of this BAPI creates a real shipment. Plan the DS4 demo accordingly.

BAPI_SHIPMENT_COST_ESTIMATE — what it actually needs

Inputs: `HEADERDATA` , `ITEMDATA` , `STAGEDATA` , deadlines, plus `DLVHEADER` and `DLVITEMS` . Exports `RETURN_COSTS` and `RETURN` — and notably **no** `TRANSPORT` , consistent with creating nothing.

NOT A DELIVERY-ONLY CALL

You cannot pass a delivery number and get a freight quote. You must construct an in-memory shipment. Any API-04 design assuming “send the DI and the transporter, get a price” is wrong about the input shape. Its non-posting behaviour is also **asserted, not proven** — we have not yet compared `VFKK` / `VFKP` key counts before and after a call.

The two BAPIs actually executed

BAPI	Input	Output	Meaning
<code>BAPI_MATERIAL_AVAILABILITY</code>	<code>15000177 / 1002 / TO</code>	<code>WM3351 Material not maintained in plant ; ATP 0.000</code>	QS4 master data absent from DS4
same	<code>MAT18 / PLQ3 / EA</code>	<code>no error; ATP 0.000</code>	Callable and authorised; DEV material has zero ATP
<code>BAPI_PO_CREATE1</code> <code>TESTRUN=X</code>	<code>POHEADER-DOC_TYPE = ZP06</code>	<code>I MMPUR_BASE 054</code> <code>E MEPO 002</code> <code>E ME 013 Document type ZP06 not allowed with doc. category F</code> <code>W W5 005</code>	ZP06 not configured in DS4

Read `ME 013` precisely. It does not say “document type does not exist”. It says the type is not *allowed for document category F* — the purchase-order category. That is a configuration statement, not a BAPI defect. `EXPPURCHASEORDER` came back blank and no commit occurred, exactly as `TESTRUN` should behave. The BAPI also echoed `ZP06` back unchanged in `EXPHEADER` — it accepted the value into the interface and rejected it at validation.

3.3 The internal function modules — the heart of the problem

Steps 2 to 5 are performed by SAP function modules in function groups `V54C` , `V54R` , `V54U` , `V54B` , `V54K` , `V54S` , `V54O` . The complete `SD_SCDS_*` family is nine modules:

Function module	Program	SAP short text (German)	Role
<code>SD_SCDS_CREATE</code>	<code>SAPLV54C</code>	Frachtkosten anlegen (online, batch)	Creates the cost document
<code>SD_SCDS_CHANGE</code>	<code>SAPLV54C</code>	—	Change
<code>SD_SCDS_RELEASE</code>	<code>SAPLV54R</code>	Überleitung der Frachtkostenpositionen	Transfer / settlement
<code>SD_SCDS_REVERSE</code>	<code>SAPLV54R</code>	Storno der Frachtkostenpositionen	Cancel / reverse
<code>SD_SCDS_SAVE</code>	<code>SAPLV54U</code>	Frachtkosten sichern	Persist
<code>SD_SCDS_SHIPMENT_UPDATE</code>	<code>SAPLV54U</code>	Fortschreiben von Transporten	Writes <code>VTTK-FBGST</code> / <code>ARGST</code>
<code>SD_SCDS_CHECK_CHANGES</code> / <code>_CHECK_COMPLETE</code>	<code>SAPLV54U</code>	—	Validation
<code>SD_SCDS_HEAD_DETERMINE_STATUS</code>	<code>SAPLV54S</code>	—	Status determination

Plus at item level: `SD_SCD_ITEM_CALCULATE` (`V54B`), `SD_SCD_ITEM_ACCT_ASSIGNMENT` (`V54K`), `SD_SCD_HISTORY_SETTLEMENT` (`V54O`), and roughly 30 further `SD_SCD_*` helpers.

THE RELEASE QUESTION, SETTLED BY SAP'S OWN NAMING

“Überleitung” is German for *transfer*, not release (“Freigabe”). `SD_SCDS_RELEASE` maps to `STABR` . So what the business calls “cost release” is technically the **transfer of freight cost items into MM** — which is exactly why completing it produces a freight purchase order and a service entry sheet.

Interfaces, and what they say about wrappability

Module	Key parameters	Coupling	Wrappable?
SD_SCDS_CREATE	I_FKART , I_PRSDT , I_BUDAT , I_OPT_COMMIT , C_REFOBJ_RANGE , E_REFOBJ_RANGE_LOCKED	No dialog dependency; own commit flag; range-based	Best candidate built for "online, batch"
SD_SCDS_RELEASE	I_REFOBJ_TAB , I_TCODE , C_SCD_TAB	No T180 ; needs document structure prepared	Moderate
SD_SCDS_SAVE	I_T180 , I_OPT_UPDATE_TASK	Requires screen-sequence control	Dialog-bound
SD_SCD_ITEM_CALCULATE	I_T180 , I_REFOBJ , I_FKART , I_OPT_CALC_TYPE	Requires screen-sequence control	Dialog-bound
SD_SCD_ITEM_ACCT_ASSIGNMENT	I_TRTYP , I_CALLING_PROGRAM , I_CUA_ROUTINE_EXT , E_FCODE	T185 / T185V screen-flow, SY-CPROG , CUA codes	Do not wrap directly
SD_SCD_HISTORY_SETTLEMENT	I_VFKP → E_SETTL_HIST	None	Clean read API for freight PO / service entry

SAP ITSELF REFUSES TO SPLIT CREATE AND SAVE

Where-used on SD_SCDS_SAVE returns LV54CU10 — which is SD_SCDS_CREATE 's own include. SD_SCDS_CREATE calls SD_SCDS_SAVE internally. Create and persist are one operation at the SAP level. Exposing them as two endpoints would invent a boundary SAP does not have.

3.4 The callability verdict

Step	Mechanism	Released	RFC	Callable today
1 · Shipment create	BAPI_SHIPMENT_CREATE	Yes	Yes	YES
— · Freight estimate	BAPI_SHIPMENT_COST_ESTIMATE	No	Yes	With risk
2 · Cost doc create	SD_SCDS_CREATE	No	No	NO
3 · Calculate	SD_SCD_ITEM_CALCULATE	No	No	NO
4 · Account assign	SD_SCD_ITEM_ACCT_ASSIGNMENT	No	No	NO
5 · Settle	SD_SCDS_RELEASE	No	No	NO
6 · PGI	OData PostGoodsIssue	Yes	—	YES

How we know — and why the method matters

- RFC status:** TFDIR-FMODE . Control: BAPI_SHIPMENT_CREATE returns R ; every SD_SCDS_* module returns blank. The control proves we are reading the right column.
- Search completeness:** patterns SD_SCD* , SD_SCDS* , *SHIPMENT_COST* , *SHIPMENTCOST* , BAPI_SHIPMENT* , V54* . An earlier narrow search using only *SHIPMENTCOST* found nothing and produced the incorrect claim "no such BAPI exists".
- Where-used completeness:** full scope, all object types selected. Control: where-used on SD_SCD_ITEM_CALCULATE returned SAP-standard includes LV54CF02 and LV54CF03 — proving the index covers standard code, so a custom-only result elsewhere is a real finding, not an index gap.

PHRASE IT AS A CLAIM ABOUT THE SEARCH

The defensible sentence is: "no shipment-cost create or release BAPI was discovered through the searches performed" — patterns and where-used both listed. **Never** "no BAPI exists".

THE CONCLUSION THAT DRIVES SECTION 6

Four of the six pre-PGI steps have no callable SAP interface. Custom ABAP development is therefore mandatory, regardless of how many endpoints we expose. This is not a design preference — it is a property of the system.

4 What we found that is not currently known

4.1 10,496 stranded cost documents

True SE16 hit counts on VFKK in QS4/700, full table:

52.4% settled		47.4% not relevant	
3,021,454 fully transferred	2,737,318 not relevant	10,496 stranded (0.18%)	
STABR	Meaning	Count	Share
C	Fully transferred	3,021,454	52.4%
blank	Not relevant for transfer	2,737,318	47.4%
A	Relevant, not transferred	10,496	0.18%
B	Partially transferred	0	—
Total		5,769,268	

STATE THE 0.18% YOURSELF, FIRST

If you present 10,496 as if it were systemic and someone divides by 5.77 million, the number turns against you and takes the rest of the briefing with it. Presented honestly it is a strong argument. Presented as a crisis it is a liability.

This 0.18% is the **best case**. It comes from ABAP running inside SAP, in the same program context, with no network between the steps, tuned over years of production. It still leaves 10,496 documents stranded. A new integration that splits these steps across a network boundary, with an orchestrator that cannot read VFKK, will not beat 0.18%.

At the 4,000 logistics calls per day cited on 17 August, 0.18% is roughly **7 stranded documents per day, about 2,600 a year** — each with calculated freight cost that never reached MM. No freight purchase order, no service entry sheet, no accrual cleared. That is a financial exposure, not a rounding error.

The traced example — know this chain cold

VFKK 1300105310 FKART Z003 STBER=C STFRE=A STABR=A

Calculated · account assignment RELEVANT but not done · transfer RELEVANT but not done

↓ VFKP item 000001 · FKPTY Z002 · 12,144.00 INR · TDLNR 0013000913 · EBELN blank · LBLNI blank

Shipment 1300105363

via VFKP-REBEL

↓ VTTP

Delivery 0080387199 LFART ZLF WBSTK=C FKSTK=C

PGI COMPLETE and invoiced — while the freight cost was never transferred

All 200 rows sampled from this population showed FKART = Z003, STBER = C, STFRE = A, STABR = A. A consistent state, not scattered failures.

4.2 Cost type is not a constant — and it is a hole in the contract

Any cost-posting call takes a shipment cost type as a **mandatory** input. Four types were observed behaving differently:

FKART	Observed lifecycle	Sample
Z001	Completes fully — all three statuses <code>C</code> ; freight PO and service entry created	Settled sample, Sep–Oct 2024
Z003	Strands — calculated, then <code>STFRE=A / STABR=A</code>	The 10,496 population
Z006	Calculates; settlement not relevant	201 rows, all 31.03.2026, one batch
Y003	Calculated only	Doc 2100005093, Feb 2024

NEW INFORMATION FOR THE ROOM

On 17 August the client described release as “a tick inside the transaction, not a separate BAPI”. Accurate for the GUI. But the **callable** form requires a cost type to be chosen and supplied — and **v1.8 §4 has no field for it**. Either it is derived from the shipment or planning point, or it becomes a caller-supplied request field. This must be decided.

4.3 Settlement is not a technical prerequisite for PGI

An open question for weeks. Now closed — and the answer is counter-intuitive.

PROVEN

SAP does not block goods issue on unsettled freight cost. Cost document `1300105310` has `STABR = A` — settlement *relevant and incomplete* — while its delivery `0080387199` shows `WBSTK = C` , PGI complete.

Why an earlier counter-example did not prove this

	First attempt	The valid one
Delivery	<code>0180915535</code>	<code>0080387199</code>
Delivery type	<code>EL</code> — not a dispatch type	<code>ZLF</code> — real dispatch, same family as the settled reference
<code>STABR</code>	blank = <i>not relevant</i>	<code>A</code> = not transferred
Proves	PGI proceeds when settlement is irrelevant — trivially true	PGI proceeds when settlement is relevant and incomplete

The distinction is the entire argument. **Do not present the first version.**

Design consequence: “release before PGI” is a **business rule, not a SAP constraint**. If CNF wants it enforced, CNF must enforce it. Equally, an orchestration that blocks waiting for settlement is imposing a constraint SAP does not impose — that should be a deliberate choice, not an inherited assumption.

Limit of the claim: it proves PGI is *possible* without settlement in this configuration. It does not prove no configuration anywhere in the landscape would block it.

4.4 This pattern is already implemented here

Where-used on `SD_SCDS_CREATE` returned **eight callers, every one custom Z code** — and no SAP standard caller. Classes `ZCLDACE_POST_UTILITY` , `ZCLDACE_WT_TRNASFER` , `ZDACE_CL_STO_PROCESS` (each with a method `SHIP_COST_CRT`); programs `ZLEIILMSDOCUMENTS_SHIPMENTCOST` , `ZLEIILMSDOCUMENTS_INTERCOMPANY` , `ZLERA2SALAUTO` , `ZLERBTSTSALAUTO` , `ZSDC013_SHIPMENT_UPLOAD_SUB` .

The closest precedent — `ZDACEFM_SHIP_COST`

Attribute	Value
Function group	<code>ZDACEFG_SHIP_COST</code> — “Shipment Cost FG”
Package	<code>ZDEL_ACE</code>
Short text	“Shipment Cost Posting FM”
Last changed	08.05.2025
Remote-enabled	No
Interface	<code>I_TKNUM</code> (mandatory) · <code>I_FKART</code> (mandatory) · <code>I_COMMIT</code> (optional)
Exports	None. No <code>RETURN</code> parameter at all.

One shipment in, cost document posted, commit under caller control. **This is the shape we are proposing, and it already exists and runs in this system** — a feasibility proof we did not have to construct.

TWO CAVEATS YOU MUST STATE, UNPROMPTED

1. This object belongs to the weighbridge programme, which is **out of C&F scope**. Cite it as evidence the pattern is proven and the shape is known — **not** as a CNF asset to reuse, and **not** as proof that CNF needs only a thin wrapper.
2. It has no **RETURN** parameter. It cannot tell its caller what it created or why it failed. That is a genuine defect for integration use, and precisely what our version must fix. **Copy the shape; do not copy the silence.**

5 Comparison with the v1.8 specification

5.1 Corrections v1.8 requires

#	v1.8 / prior belief	Runtime evidence	Impact
1	<code>VFKK-STFRE</code> is the cost release status	It is account assignment (<code>STFRE_K</code>); blank = “not relevant”	High
2	<code>FKNUM = REBEL = TKNUM</code>	False. Join is <code>VFKP-REBEL</code> → <code>VTTK-TKNUM</code>	High
3	Release evidence is <code>STFRE</code> / <code>STABR</code> alone	Add <code>VTTK-FBGST</code> / <code>ARGST</code> and <code>VFKP-EBELN</code> / <code>LBLNI</code>	Medium
4	<code>PickAndBatchSplitOneItem</code> has 4 parameters	5 — <code>SplitQuantityUnit</code> missing	Medium call would fail
5	<code>CreateBatchSplitItem</code> unspecified	6, including <code>PickQuantityInSalesUOM</code>	Medium
6	<code>GoodsMovementCode</code> unknown	<code>T158G</code> has 7 values; <code>01</code> = MB01 goods receipt for PO — candidate pending a run	Medium
7	Billing creatable via <code>API_BILLING_DOCUMENT_SRV</code>	Impossible — all 8 entity sets <code>creatable=false</code>	High
8	Stage-B transport fields have no home	Append fields on <code>BAPISHIPMENTHEADER</code>	High unblocks \$4
9	Cost type not mentioned	<code>FKART</code> is a mandatory input and varies by scenario	High missing field
10	Stage B route “unidentified”	Identified: <code>SD_SCDS_CREATE</code> → <code>SD_SCDS_RELEASE</code> , neither callable	High

5.2 The gap in v1.8 §4

API-03 §4 (Request properties) has field-level rows for stages A, C, D and E — and **nothing for stage B**, only a prose note that the route is unidentified. §8 parks the `ZZ*` transport fields as belonging to that unidentified route.

Both are now fixable. The stage-B request contract is `BAPISHIPMENTHEADER` (including the `ZZ*` fields) plus `ITEMDATA` referencing the delivery, plus a decision on how `FKART` is supplied. This is the concrete next deliverable — produced as a delta report, **not** by editing the v1.8 workbook directly.

5.3 API-by-API status

Counting rule: an operation is validated only when a representative input was executed, the raw response captured, and the result reconciled to an SAP document. **On that rule, two operations are validated.**

API	Purpose	Mechanism	Status
API-01	Goods receipt (MIGO)	<code>API_MATERIAL_DOCUMENT_SRV</code> deep insert	Blocked — no valued material; <code>GoodsMovementCode</code> candidate <code>01</code>
API-02	Create delivery	<code>API_OUTBOUND_DELIVERY_SRV</code> ; v=2	Blocked — no predecessor documents
API-03	Invoice chain	Composite — section 6	Pre-PGI half requires custom development
API-04	Freight estimate	<code>BAPI_SHIPMENT_COST_ESTIMATE</code>	Not released; in-memory shipment required; non-posting unproven
API-05	Stock availability	<code>API_MATERIAL_STOCK_SRV</code>	Contract correct and read-only; no stock in DS4
API-06 / 07	E-way / correction	DigiGST / EY eDocument	Existing route; boundary not traced
API-08	STO PO read	<code>API_PURCHASEORDER_PROCESS_SRV</code>	Service works; no ZP06 data in DS4
API-09	STO delivery read	<code>API_OUTBOUND_DELIVERY_SRV</code> ; v=2	Service works; ZNL absent in DS4
API-10	STO invoice read	<code>API_BILLING_DOCUMENT_SRV</code>	<code>SalesDocument</code> exists; VBRP-AUBEL mapping unproven
API-11	Create STO PO	<code>BAPI_PO_CREATE1</code> / OData	ZP06 rejected — <code>ME 013</code> configuration gap
API-12	Update DI quantity	PATCH <code>A_OutbDeliveryItem</code>	Quantity updatable; unit is not ; ETag untested

6 The decision: one pre-PGI API

6.1 The question, stated precisely

Scope is **shipment creation** → **freight estimate** → **cost document creation and calculation** → **cost release**, ending immediately before PGI. Billing, e-invoice, e-way and the Hybris repair cron are outside this decision.

The question is whether CNF calls SAP **once** for that span, or **three times**.

6.2 Why one — four arguments

1 · Custom development is required either way

Four of six steps have no callable interface. Three endpoints means three custom wrappers, not fewer. The choice is only how many doors we cut in the wall, not whether we cut any. **“Use standard APIs instead” is not an available option** — and we can now prove that rather than assert it.

2 · SAP has already merged two of the steps

`SD_SCDS_CREATE` calls `SD_SCDS_SAVE` internally. Splitting create from persist would invent a boundary SAP does not have. This also matches what the room agreed on 17 August: costing-document creation and costing release merged into one call, reducing four calls to three.

3 · State lives in SAP, so the caller cannot own it

Chain state is in `VTTK`, `VFKK` / `VFKP` and `LIKP`. Only SAP can answer “how far did this get”. Three calls means CPI or the portal holding a state machine for data it cannot read. One call that reads those documents and continues from where it finds itself is **naturally idempotent** — the single most valuable property this interface can have.

4 · The 10,496

Splitting steps is survivable; SAP's own implementation survives it at 0.18%. But nothing about three endpoints makes that number smaller, and two things make it larger: a network boundary between the steps, and an orchestrator that cannot see the cost tables.

6.3 What the API must return

Modelled on the real completed chain in 2.2:

```
{ "shipment":      "1100605471",
  "costDocument":  "1100608871",
  "costType":      "Z001",
  "calculated":    true,           // VFKP-STBER = C
  "accountAssigned": true,        // VFKP-STFRE = C
  "settled":       true,          // VFKP-STABR = C
  "freightPO":     "6000105831",  // VFKP-EBELN
  "serviceEntry":  "1003382363",  // VFKP-LBLNI
  "stage":         "READY_FOR_PGI",
  "messages":      [ SAP message class + number + text ] }
```

Stage-level status, not a boolean. Every step that completed is named with the document it produced. A failure returns the step it stopped at plus SAP's own message — exactly what the two-class error taxonomy agreed on 17 August needs to route functional versus technical errors.

6.4 Three conditions to state alongside it

1. **It is not atomic, and cannot be.** Once the shipment commits, it is committed. Recovery is reversal, not rollback. Equally true of one API or three — so not an argument between them, but it will be held against the design later if nobody wrote it down now. The word “rollback” appeared in the 17 August readout; **withdraw it before it is tested**.
2. **It is resumable, not fire-and-forget.** Re-invoking reads `VTTK` / `VFKK` / `VFKP`, determines what already exists, and continues. This is what prevents a duplicate shipment and what prevents another 10,496.
3. **The SAP wrapper does not own process state.** No process id, no stage ledger, no replay store in ABAP. SAP holds no cross-step process state and cannot roll back a committed PGI however the ABAP is written. Process orchestration belongs to the orchestrator — which the 17 August meeting placed in Hybris.

ON NAMING

The business stopped the `ZCNF_*` custom OData portfolio on 14 August, and v1.8 was rebuilt to drop those names. The engineering here is justified and the evidence supports it — but frame it as **exposing internal SAP function modules that have no released interface**, which is narrow and defensible. Presenting a new custom service tree reopens a decision that already went against the project.

7 What remains open

7.1 Open items, ranked

#	Question	Why it matters	How to close it
1	Does <code>BAPI_SHIPMENT_CREATE</code> trigger cost-document creation on save?	If yes, the exposed surface shrinks materially	Inspect <code>SAPLV56I_BAPI</code> ; where-used is evidence against but does not exclude a dynamic call
2	Is <code>BAPI_SHIPMENT_COST_ESTIMATE</code> genuinely non-posting?	API-04 depends on it creating nothing	<code>VFKK</code> / <code>VFKP</code> key counts before and after a call
3	What invokes <code>SD_SCDS_RELEASE</code> ?	Completes the settlement call path	Zero static callers at full scope — needs a runtime trace during a real VI02 settlement
4	How is <code>FKART</code> determined?	Mandatory input, currently unspecified	Functional decision plus configuration read
5	Function-import parameter encoding	Affects every delivery v2 call	One executed call settles it

7.2 The DS4 blocker

DS4/200 cannot currently run a single pre-PGI evidence run. Configuration is present; master and transaction data are not.

Object	Table	Rows	Blocks
Vendors / forwarding agents	<code>LFA1</code>	0	Everything from shipment onward — no transporter can be entered
Storage-location stock	<code>MARD</code>	0	Picking, PGI
Material valuation	<code>MBEW</code>	0	Any goods movement
Batches	<code>MCHA</code>	0	Batch split
Material/plant views	<code>MARC</code>	2	<code>MAT18</code> , <code>MAT19</code> , both plant <code>PLQ3</code> — STO needs one material in two plants
Sales orders / deliveries	<code>VBAK</code> / <code>LIKP</code>	0	All predecessors

Present and usable: `TVTK` 7 shipment types · `TTDS` 30+ planning points · `TVR0` 25+ routes · `TVFT` 6 cost document types · `T161` 35 PO types including standard `UB` / `UB2` .

CNF-specific types are **absent**, each proven with a positive control so the filter itself is trustworthy: `ZP06` (rejected by BAPI with `ME 013`), `ZNL` (control `LF` returned exactly 1 row), `ZSTO` (control `F2` returned exactly 1 row).

The decision this forces

	Option A — generic	Option B — CNF parity
Uses	Standard <code>UB</code> , standard delivery and billing types	<code>ZP06</code> , <code>ZNL</code> , <code>ZSTO</code> transported into DS4
Proves	API mechanics, payload shape, commit boundaries, retry	Actual CNF behaviour and derivations
Does not prove	Anything about <code>ZP06/ZNL/ZSTO</code> determination	—
Speed	Fast once master data exists	Requires a customizing transport

Recommendation: A now, B before sign-off. Option A unblocks most steps immediately and is reversible. Option B is required before anyone signs a contract naming `ZP06`, `ZNL` or `ZSTO`. Option A results must never be presented as CNF configuration parity.

7.3 What we are asking for

The minimum disposable test chain in DS4/200: one valued material with stock in a named plant and storage location; a batch-managed variant with two batches; a customer and a sales order or STO; one delivery not yet PGI-complete; **at least one vendor with partner function `SP`** ; a route; freight condition records so a real rate returns; and confirmation of which shipment type, planning point and cost document type to use.

Plus one process answer no technical inspection can produce: **which transaction or job the business actually runs to create and settle shipment costs.**

Full field-level list: `PRE_PGI_EVIDENCE_RUN_BLOCKER_PACKET.md` , 15 items.

8 Quick reference

8.1 Document numbers to know cold

Number	What	Use it to show
9004953084	Delivery, ZLF , 17.08.2026	The complete reference chain
1100605471	Shipment	Created 11:11:28
1100608871	Cost document, Z001	Settled 11:11:30
6000105831 / 1003382363	Freight PO / service entry	What settlement actually produces
4918168265	PGI material doc, movement 601	11:11:31 — one second after settlement
1300105310	Cost document, Z003 , STABR=A	The stranded example
0080387199	Delivery, ZLF , WBSTK=C	PGI'd despite unsettled freight
2100005093	Cost document, Y003	Calculated, never settled; REBEL ≠ FKNUM

8.2 Sentences to use verbatim

- “Four of the six pre-PGI steps have no callable SAP interface. Custom development is mandatory either way.”
- “ VFKK-STFRE is account assignment, not release. Blank means not relevant, not not-done.”
- “Cost release is technically the transfer of freight cost into MM — that is why it produces a purchase order and a service entry sheet.”
- “0.18% is the best case, from code running inside SAP with no network between the steps.”
- “SAP does not block PGI on unsettled freight. Release before PGI is your business rule, not SAP's constraint.”
- “No shipment-cost create or release BAPI was discovered through the searches performed.”

8.3 Traps — things not to say

Do not say	Because
“with rollback”	Post-commit there is no rollback, only reversal
10,496 presented as systemic	It is 0.18% — state that yourself, first
“the existing class proves CNF needs only a wrapper”	It is out-of-scope precedent, nothing more
“released” for a remote-enabled module	Different attributes; three here are RFC but not released
“validated” on metadata or an empty 200	Neither proves business behaviour
“no BAPI exists”	Say “none discovered through the searches performed”

Evidence base: [sessions/2026-08-18-runtime-certification/](#) — [stageb-qs4/](#) document traces and status counts · [stageb-route/](#) where-used results and captured interfaces · [evidence/](#) OData metadata and read responses · [analysis/](#) parsed runtime indexes.